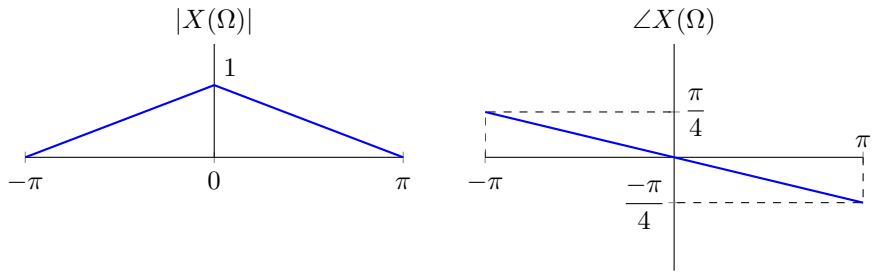
Middle East Technical University
Department of Electrical and Electronics Engineering
EE301: Signals and Systems I

Solutions for Homework 5
January 7, 2018
If you face any problem or mistake please contact Ömer Çayır, *ocayir@metu.edu.tr*, DZ-10.

1. (a) The Fourier transform of the DT signal $x[n]$ is

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

The magnitude and phase of $X(\Omega)$ are shown below for $|\Omega| < \pi$.



Let $X_a(\Omega)$ be the DTFT of $x[-n]$.

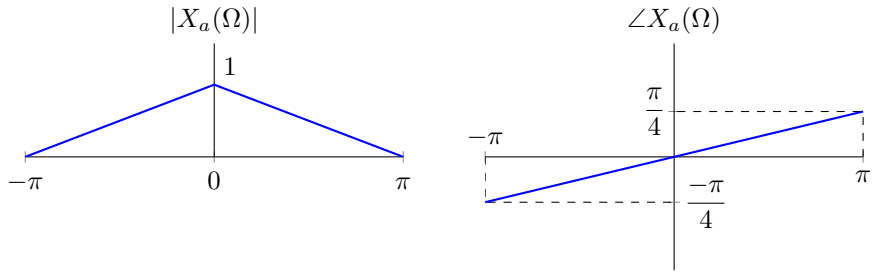
$$\begin{aligned} X_a(\Omega) &= \sum_{n=-\infty}^{\infty} x[-n]e^{-j\Omega n} \\ &= \sum_{m=-\infty}^{\infty} x[m]e^{j\Omega m} \qquad (\text{for } n = -m) \\ &= X(-\Omega) \end{aligned}$$

The magnitude and phase of $X_a(\Omega)$ are found as

$$|X_a(\Omega)| = |X(-\Omega)|$$

$$\angle X_a(\Omega) = \angle X(-\Omega)$$

Over the range $-\pi < \Omega < \pi$, they are shown below.



(b) Let $X_b(\Omega)$ be the DTFT of $x[n-1]$.

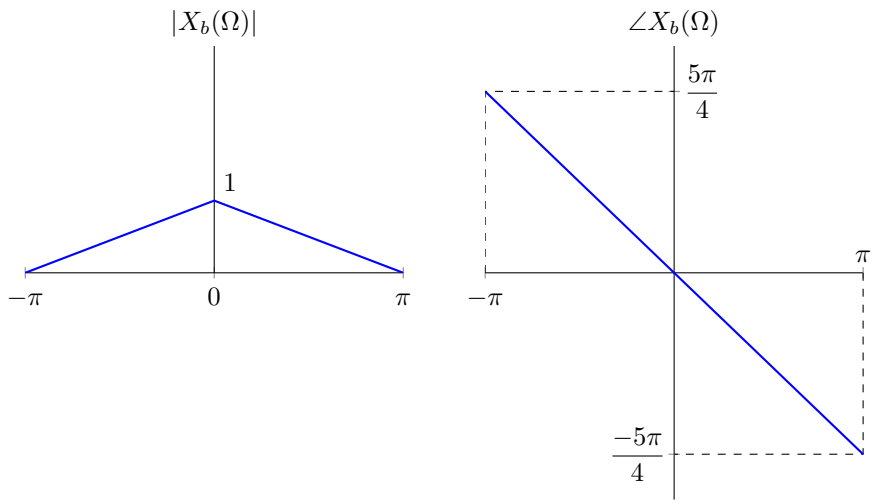
$$\begin{aligned} X_b(\Omega) &= \sum_{n=-\infty}^{\infty} x[n-1]e^{-j\Omega n} \\ &= \sum_{m=-\infty}^{\infty} x[m]e^{-j\Omega(m+1)} \qquad \text{(for } n = m+1\text{)} \\ &= e^{-j\Omega} \sum_{m=-\infty}^{\infty} x[m]e^{-j\Omega m} \\ &= e^{-j\Omega} X(\Omega) \end{aligned}$$

The magnitude and phase of $X_b(\Omega)$ are found as

$$|X_b(\Omega)| = |X(\Omega)|$$

$$\angle X_b(\Omega) = -\Omega + \angle X(\Omega)$$

Over the range $-\pi < \Omega < \pi$, they are shown below.



2. (a) The frequency response $H(\Omega)$ is

$$H(\Omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\Omega n}$$

The impulse response values of the new filter, $g[n]$, are related to $h[n]$ as follows:

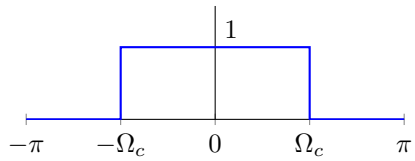
$$g[n] = \begin{cases} -h[n], & \text{if } n \neq 0 \\ 1 - h[0], & \text{if } n = 0 \end{cases}$$

The frequency response of the new filter, $G(\Omega)$, can be written as

$$
\begin{aligned}
G(\Omega) &= \sum_{n=-\infty}^{\infty} g[n]e^{-j\Omega n} \\
&= g[0] + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} g[n]e^{-j\Omega n} \\
&= 1 - h[0] - \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} h[n]e^{-j\Omega n} \\
&= 1 - h[0] - \Big(H(\Omega) - h[0]\Big) \\
&= 1 - H(\Omega)
\end{aligned}
$$

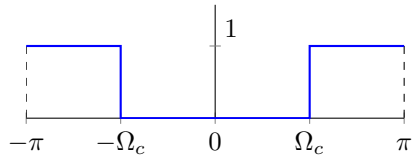
The original filter with impulse response $h[n]$ is given to be a low-pass filter with unit gain, i.e. $|H(0)| = 1$. Let $h[n]$ be an ideal low-pass filter with the frequency response shown below.

$$|H(\Omega)| = H(\Omega)$$



Then, the new impulse response $g[n]$ is a high-pass filter with the frequency response shown below.

$$|G(\Omega)| = 1 - H(\Omega)$$



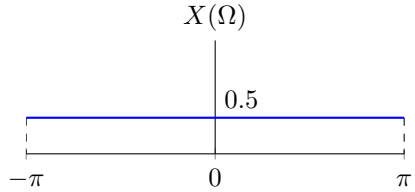
(b)

$$
\begin{aligned}
z[n] = \frac{1}{2}\operatorname{sinc}\left(\frac{n}{2}\right) &\overset{\mathcal{F}}{\longleftrightarrow} Z(\Omega) = \operatorname{rect}\left(\frac{\Omega}{\pi}\right) \\
w[n] = \cos\left(\frac{\pi}{2}n\right) &\overset{\mathcal{F}}{\longleftrightarrow} W(\Omega) = \pi \sum_{\ell=-\infty}^{\infty} \left\{\delta\left(\Omega - \frac{\pi}{2} - 2\pi\ell\right) + \delta\left(\Omega + \frac{\pi}{2} - 2\pi\ell\right)\right\} \\
x[n] = z[n]w[n] &\overset{\mathcal{F}}{\longleftrightarrow} X(\Omega) = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} Z(\Omega-\theta)W(\theta)\,\mathrm{d}\theta
\end{aligned}
$$

$$
\begin{aligned}
X(\Omega) &= \frac{1}{2}\int_{-\pi}^{\pi} \operatorname{rect}\left(\frac{\Omega-\theta}{\pi}\right) \underbrace{\sum_{\ell=-\infty}^{\infty} \left[\delta\left(\theta - \frac{\pi}{2} - 2\pi\ell\right) + \delta\left(\theta + \frac{\pi}{2} - 2\pi\ell\right)\right]}_{\delta\left(\theta-\frac{\pi}{2}\right)+\delta\left(\theta+\frac{\pi}{2}\right) \text{ for } \theta\in[-\pi,\pi]} \mathrm{d}\theta \\
&= \frac{1}{2}\operatorname{rect}\left(\frac{\Omega-\frac{\pi}{2}}{\pi}\right) + \frac{1}{2}\operatorname{rect}\left(\frac{\Omega+\frac{\pi}{2}}{\pi}\right)
\end{aligned}
$$

The spectrum $X(\Omega)$ is shown below.

$$X(\Omega)$$



The output signal $y[n]$ is

$$\begin{aligned} y[n] &= h[n] * x[n] \\ &= \mathcal{F}^{-1}\{H(\Omega)X(\Omega)\} \\ &= \frac{1}{2}\mathcal{F}^{-1}\{H(\Omega)\} \qquad \text{(remember that } X(\Omega) = 0.5 \text{ for } |\Omega| < \pi) \\ &= \frac{1}{2}h[n] \\ &= \frac{1}{4}\operatorname{sinc}\left(\frac{n}{2}\right) \end{aligned}$$

3. If $x[n]$ is periodic, there exists a positive integer N such that $x[n+N] = x[n]$ and its fundamental period is the smallest nonzero period N satisfying this relation.

(a) The discrete-time signals are given as follows:

$$x_1[n] = 2.5\cos(n\pi/3 + 2.5)$$

$$x_2[n] = 3\sin(4n/5 - 1) + 3$$

For $x_1[n]$,

$$\frac{\Omega_0}{2\pi} = \frac{\pi/3}{2\pi} = \frac{1}{6}$$

is rational. Hence, $x_1[n]$ is **periodic**. Let's find N satisfying $x_1[n+N] = x_1[n]$.

$$2.5\cos\left(\frac{\pi}{3}(n+N) + 2.5\right) = 2.5\cos\left(\frac{\pi}{3}n + 2.5\right) \implies \frac{\pi}{3}N = 2\pi k \text{ for } k \in \mathbb{Z}$$

Then, we obtain $N = 6k$, and the fundamental period of $x_1[n]$ is $N = 6$ (for $k = 1$).

The DTFS representation of $x_1[n]$ is

$$x_1[n] = \sum_{k=\langle N\rangle} a_k e^{jk\frac{2\pi}{N}n} = \sum_{k=k_0}^{k_0+N-1} a_k e^{jk\frac{\pi}{3}n}$$

To find a_k, we perform some algebraic manipulations on given expression

$$\begin{aligned} x_1[n] &= 2.5\cos\left(\tfrac{\pi}{3}n + 2.5\right) \\ &= 2.5\frac{e^{j\left(\frac{\pi}{3}n+2.5\right)} + e^{-j\left(\frac{\pi}{3}n+2.5\right)}}{2} \\ &= 1.25e^{j2.5}e^{j\frac{\pi}{3}n} + 1.25e^{-j2.5}e^{-j\frac{\pi}{3}n} \end{aligned}$$

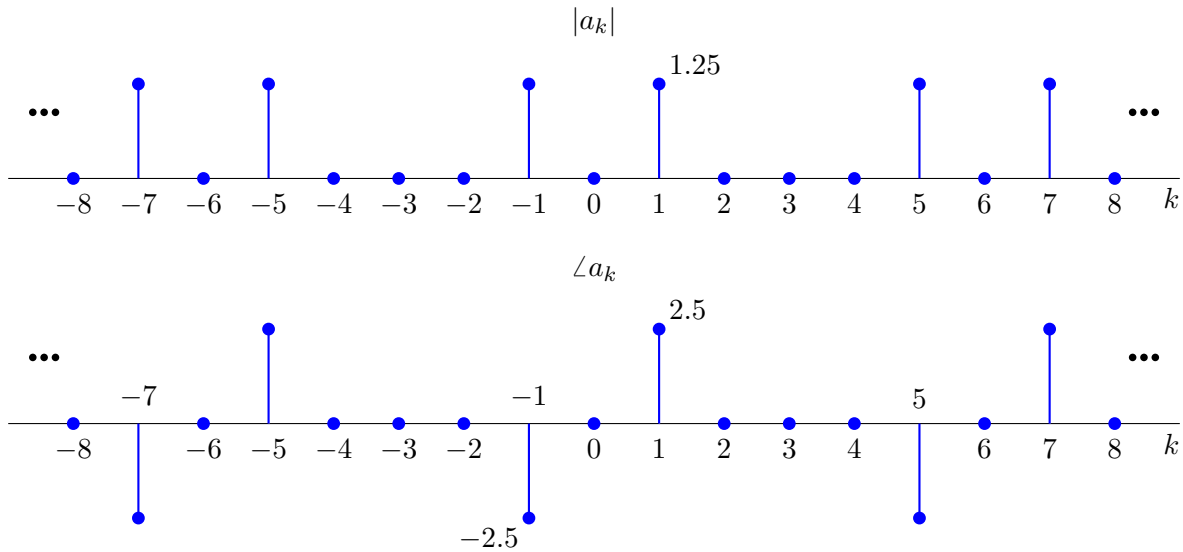
Then, we replace LHS of expression above with the DTFS representation of $x_1[n]$ for $k_0 = -2$

$$\sum_{k=-2}^{3} a_k e^{jk\frac{\pi}{3}n} = 1.25e^{j2.5}e^{j\frac{\pi}{3}n} + 1.25e^{-j2.5}e^{-j\frac{\pi}{3}n}$$

and obtain the DTFS coefficients

$$a_k = \begin{cases} 1.25e^{j2.5}, & k = 1 + 6\ell \text{ for } \ell \in \mathbb{Z} \\ 1.25e^{-j2.5}, & k = -1 + 6\ell \text{ for } \ell \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}$$

The magnitude and phase of a_k are shown below.



For $x_2[n]$,

$$\frac{\Omega_0}{2\pi} = \frac{4/5}{2\pi} = \frac{2}{5\pi}$$

is irrational. Hence, $x_2[n]$ is **not periodic**.

(b) The signal $x[n]$ is given as a periodic signal with fundamental period N and DTFS coefficients a_k.

i. After performing some algebraic manipulations on given expression, we obtain

$$x[n]\cos(\pi n) = x[n]\left(\frac{e^{j\pi n} + e^{-j\pi n}}{2}\right) = x[n]\left(\frac{e^{j\frac{N}{2}(\frac{2\pi}{N})n} + e^{-j\frac{N}{2}(\frac{2\pi}{N})n}}{2}\right)$$

To express the DTFS coefficients of the given signal in terms of a_k, we can use the DTFS properties as follows:

$$\begin{aligned}
x[n] &\overset{\text{DTFS}}{\longleftrightarrow} a_k \\
e^{j\frac{N}{2}(\frac{2\pi}{N})n}x[n] &\overset{\text{DTFS}}{\longleftrightarrow} a_{k-\frac{N}{2}} && \text{(Frequency shifting)} \\
\frac{1}{2}e^{j\frac{N}{2}(\frac{2\pi}{N})n}x[n] &\overset{\text{DTFS}}{\longleftrightarrow} \frac{1}{2}a_{k-\frac{N}{2}} && \text{(Linearity)} \\
\frac{1}{2}e^{j\frac{N}{2}(\frac{2\pi}{N})n}x[n] + \frac{1}{2}e^{-j\frac{N}{2}(\frac{2\pi}{N})n}x[n] &\overset{\text{DTFS}}{\longleftrightarrow} \frac{1}{2}a_{k-\frac{N}{2}} + \frac{1}{2}a_{k+\frac{N}{2}} && \text{(Linearity)} \\
x[n]\cos(\pi n) &\overset{\text{DTFS}}{\longleftrightarrow} b_k = \frac{1}{2}a_{k-\frac{N}{2}} + \frac{1}{2}a_{k+\frac{N}{2}}
\end{aligned}$$

The DTFS representation of the given signal is

$$x[n]\cos(\pi n) = \sum_{k=\langle N\rangle} b_k e^{jk\frac{2\pi}{N}n} = \sum_{k=0}^{N-1}\left(\frac{1}{2}a_{k-\frac{N}{2}} + \frac{1}{2}a_{k+\frac{N}{2}}\right)e^{jk\frac{2\pi}{N}n}$$

ii. The signal is given as

$$y[n] = \begin{cases} 0, & n \text{ even} \\ x[n], & n \text{ odd} \end{cases}$$

It can be expressed as

$$y[n] = x[n]w[n],$$

where

$$w[n] = \frac{1-\cos(\pi n)}{2} \begin{cases} 0, & n \text{ even} \\ 1, & n \text{ odd} \end{cases}$$

Then, we get

$$y[n] = \frac{1}{2}x[n] - \frac{1}{2}x[n]\cos(\pi n)$$

By using the result of part (i), the DTFS coefficients of the given signal in terms of a_k are

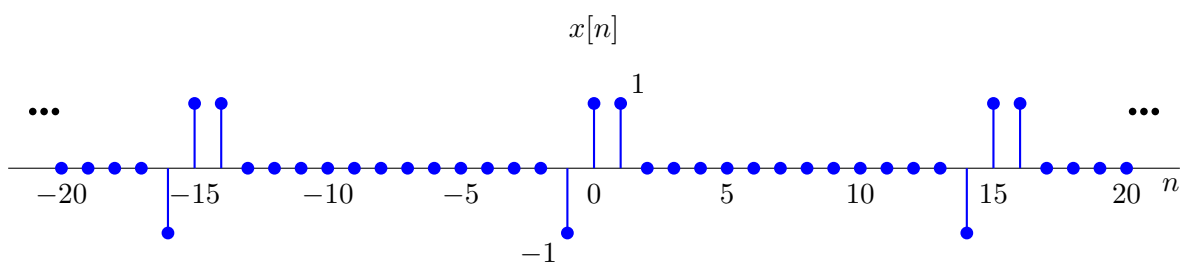
$$y[n] \overset{\text{DTFS}}{\longleftrightarrow} c_k = \frac{1}{2}a_k - \frac{1}{2}b_k = \frac{1}{2}a_k - \frac{1}{4}a_{k-\frac{N}{2}} - \frac{1}{4}a_{k+\frac{N}{2}}$$

The DTFS representation of the given signal is

$$y[n] = \sum_{k=\langle N\rangle} c_k e^{jk\frac{2\pi}{N}n} = \sum_{k=0}^{N-1}\left(\frac{1}{2}a_k - \frac{1}{4}a_{k-\frac{N}{2}} - \frac{1}{4}a_{k+\frac{N}{2}}\right)e^{jk\frac{2\pi}{N}n}$$

4. (a) A periodic signal with fundamental period $N = 15$ is given as

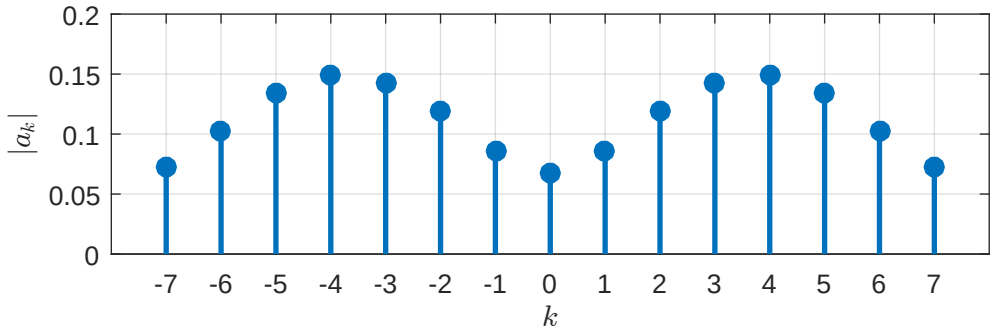
$$x[n] = \begin{cases} 1, & n = 0, 1 \\ -1, & n = 14 \\ 0, & 2 \le n \le 13 \end{cases}$$

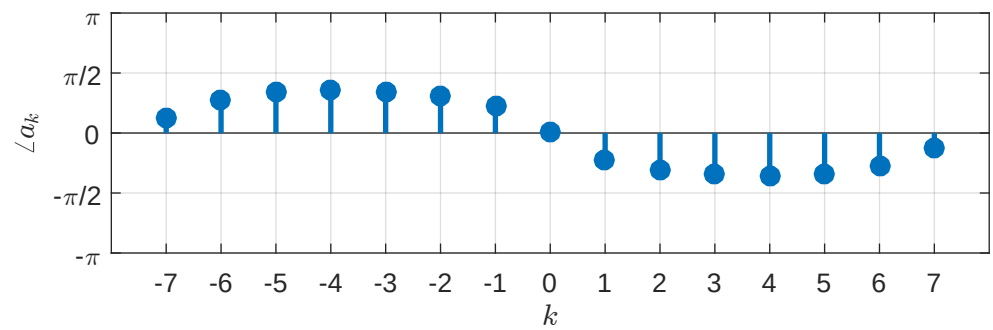


(b) The DTFS coefficients of $x[n]$ are

$$\begin{aligned} a_k &= \frac{1}{N}\sum_{n=\langle N\rangle} x[n]e^{-jk\frac{2\pi}{N}n} \\ &= \frac{1}{15}\sum_{n=-7}^{7} x[n]e^{-jk\frac{2\pi}{15}n} \\ &= \frac{1}{15}\left(-e^{jk\frac{2\pi}{15}} + 1 + e^{-jk\frac{2\pi}{15}}\right) \\ &= \frac{1}{15}\left[1 - j2\sin\left(k\frac{2\pi}{15}\right)\right] \end{aligned}$$

(c) The magnitude and phase of a_k are shown below.

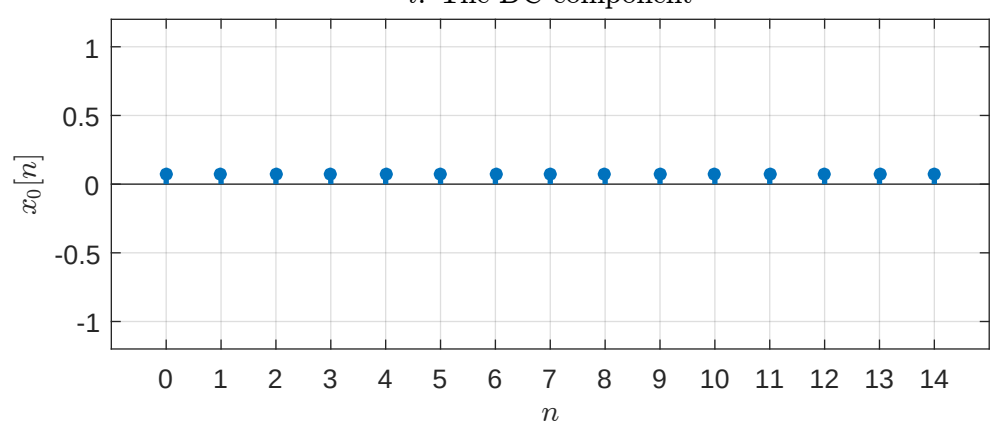




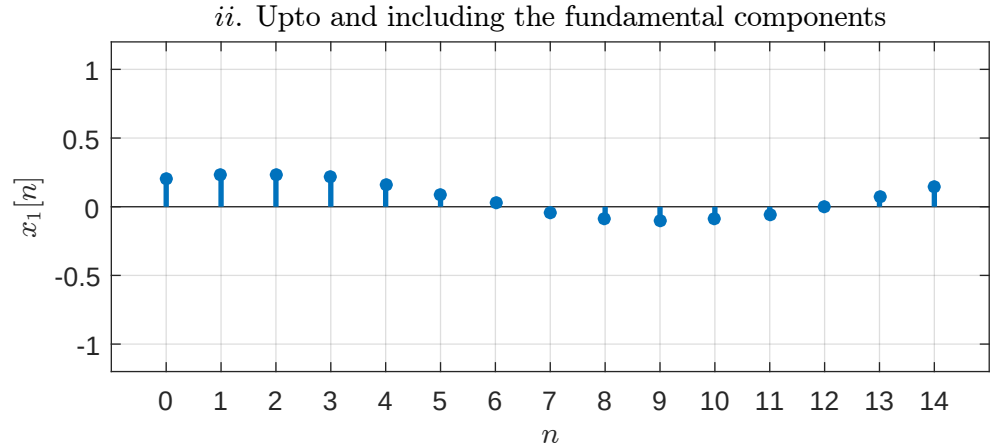
(d) The approximation to $x[n]$ is given as follows:

$$x_M[n] = \sum_{n=-M}^{M} a_k e^{jk(2\pi/N)n}$$

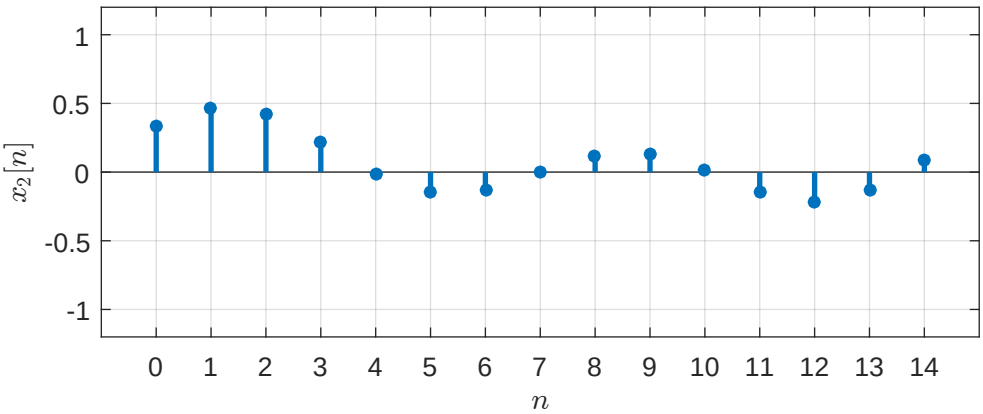
i. The DC component



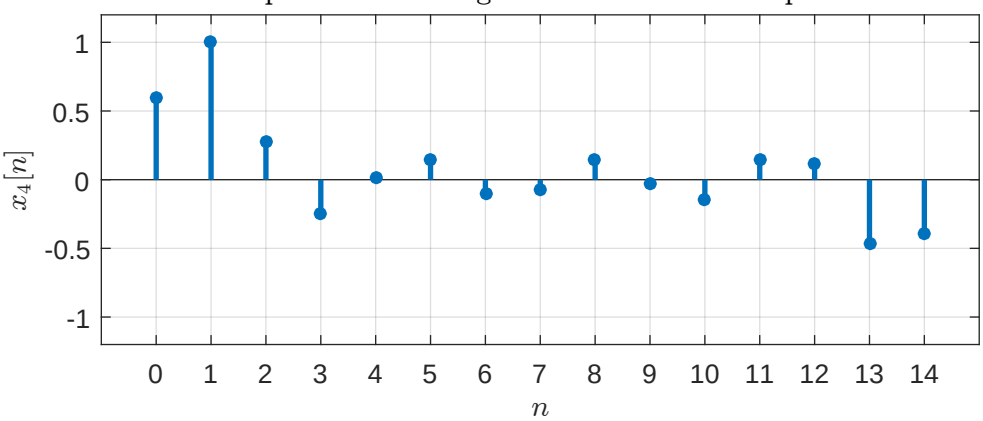
ii. Upto and including the fundamental components



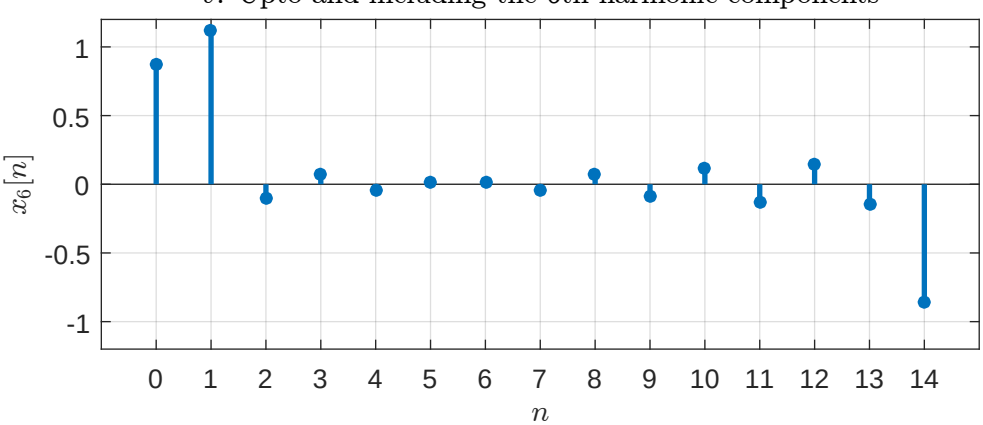
iii. Upto and including the 2nd harmonic components



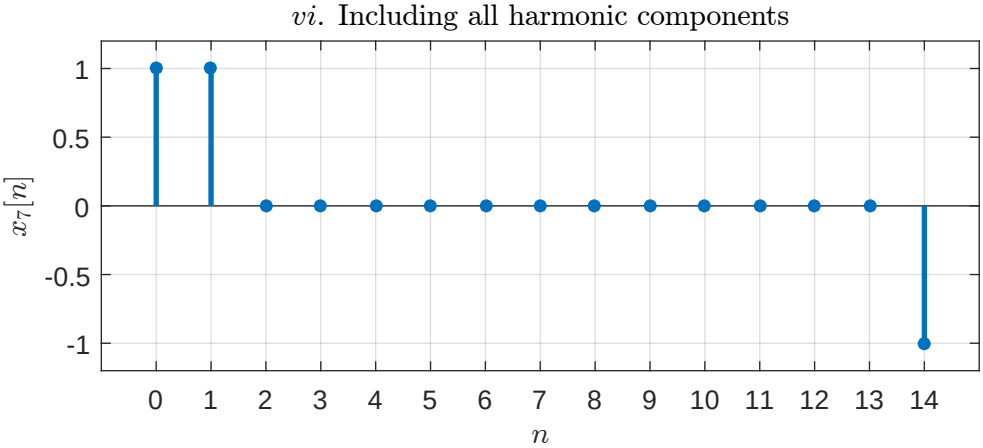
iv. Upto and including the 4th harmonic components



v. Upto and including the 6th harmonic components



vi. Including all harmonic components



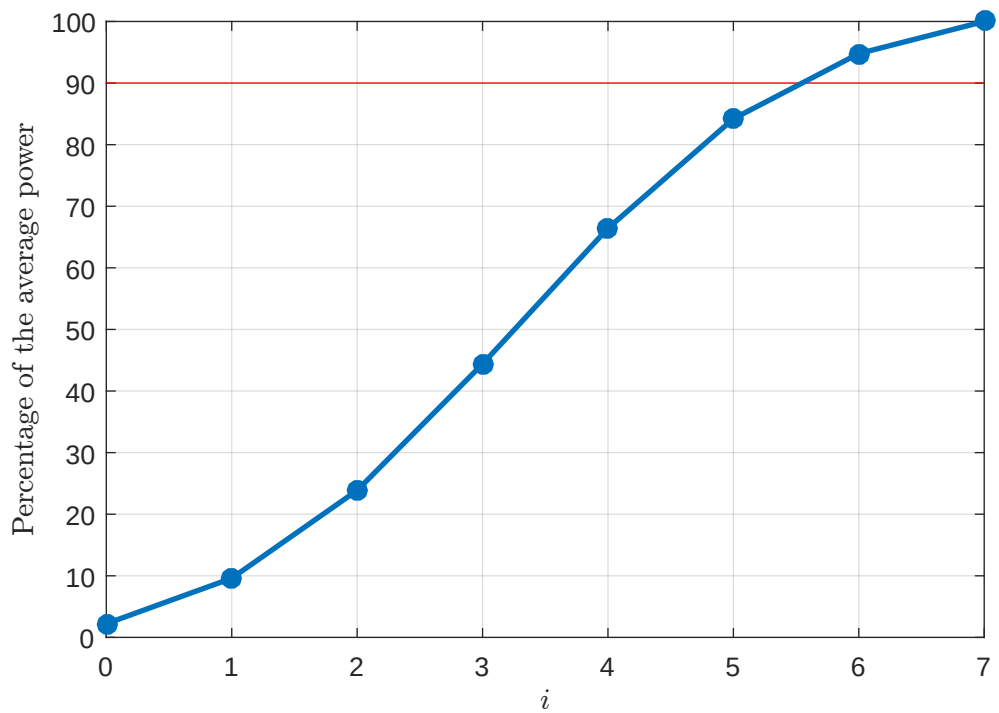
The more harmonic components are added, the more accurate the approximation becomes.

(e) The Parseval's relation for DT periodic signals is given as

$$\frac{1}{N}\sum_{n=\langle N\rangle}|x[n]|^2 = \sum_{k=\langle N\rangle}|a_k|^2$$

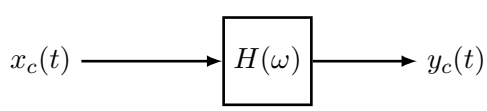
The average power of the periodic signal $x[n]$ is

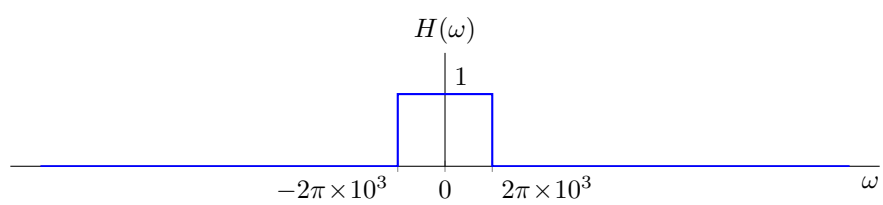
$$\frac{1}{15}\sum_{n=0}^{14}|x[n]|^2 = \frac{3}{15} = 0.2$$

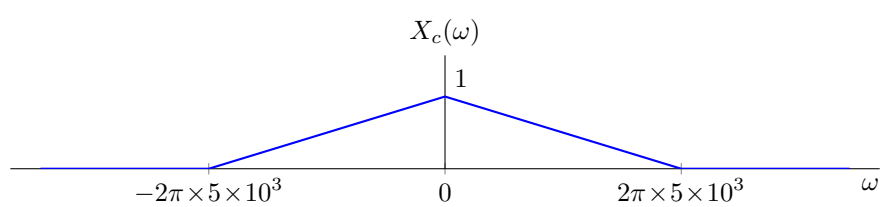


The computed value exceeds 90% of the average power for $i = 6$.

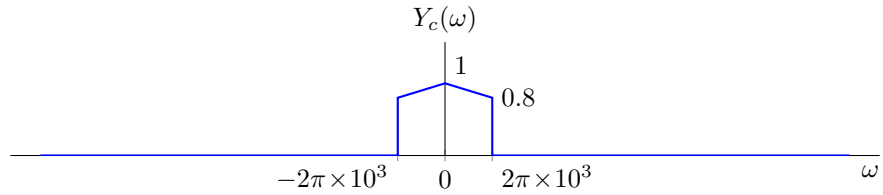
5. (a)







$$Y_c(\omega) = X_c(\omega)H(\omega) = \begin{cases} X_c(\omega), & |\omega| \le 2\pi\times10^3 \\ 0, & \text{otherwise} \end{cases}$$



(b) We should find the Nyquist rate to determine the range of sampling frequencies ω_s and sampling periods T that will allow perfect recovery of $x_c(t)$ from its samples $x[n]$.

We know that $X_c(\omega) = 0$ for $|\omega| > \omega_M = 2\pi\times5\times10^3$. Then, the Nyquist rate is

$$2\omega_M = 2\pi\times10\times10^3$$

The range of sampling frequencies is

$$\omega_s > 2\omega_M = 20\pi\times10^3$$

and the range of sampling periods is

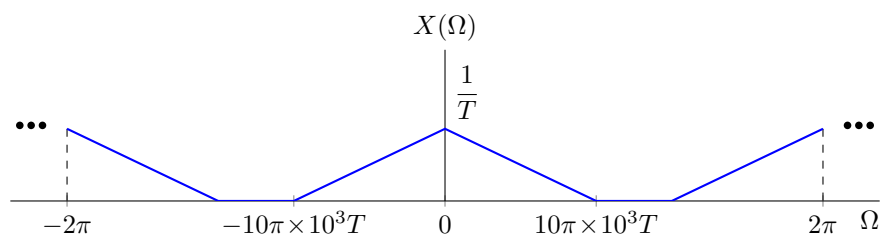
$$T = \frac{2\pi}{\omega_s} \implies T < \frac{1}{10}\times10^{-3} \text{ sec.}$$

(c) For the sampling period T, the DT signal at the input of $G(\Omega)$ is

$$x[n] = x_c(nT)$$

and its spectrum is

$$x[n] \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad X(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c((\Omega - 2\pi k)/T)$$

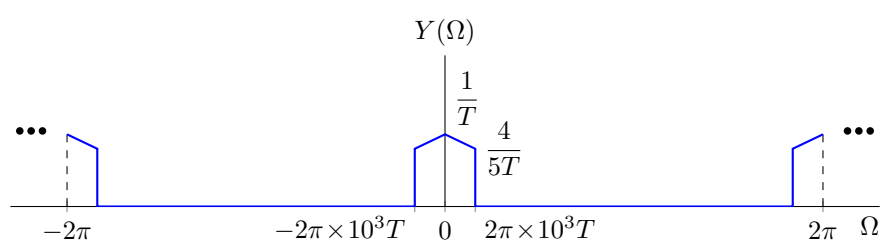


Similarly, the DT signal at the output of $G(\Omega)$ is

$$y[n] = y_c(nT)$$

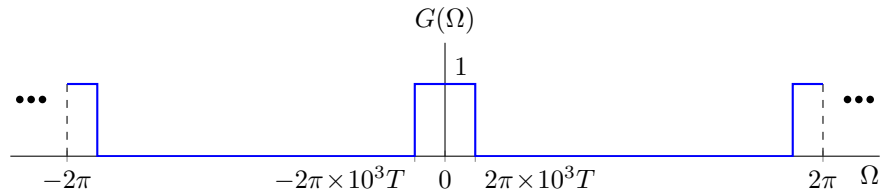
and its spectrum is

$$y[n] \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad Y(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} Y_c((\Omega - 2\pi k)/T)$$



(d) To obtain the desired $y_c(t)$ using DT processing, the required DT filter is

$$G(\Omega) = H(\Omega/T), \qquad |\Omega| < \pi$$



(e) The given CT processing scheme involves a band-limited input signal $x_c(t)$, i.e. $X_c(\omega) = 0$ for $|\omega| > \omega_M = 2\pi\times5\times10^3$, and a low-pass filter with the cut-off frequency $\omega_c = 2\pi\times10^3$. Owing to $\omega_c < \omega_M$, it is possible to obtain the spectrum of $y[n]$ without aliasing and the desired $y_c(t)$ using DT processing even if the sampling period does not satisfy the Nyquist sampling theorem. However, there will be aliasing in the spectrum of $x[n]$, when $\omega_s < 2\omega_M$.

For the sampling with the sub Nyquist rate, there are two cases as follows:

1. If $2\omega_M > \omega_s > \omega_M + \omega_c$, then the perfect reconstruction of $y_c(t)$ is **possible**.
2. If $\omega_s < \omega_M + \omega_c$, then the perfect reconstruction of $y_c(t)$ is **not possible**.

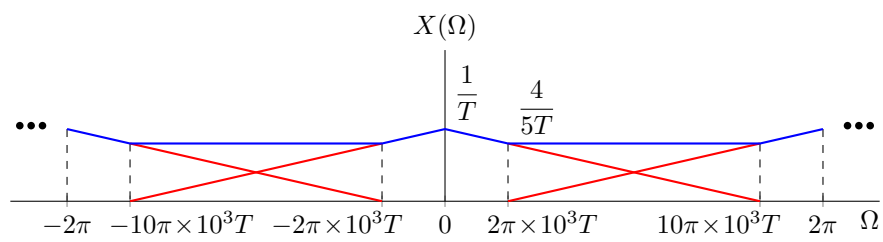
When $2\omega_M > \omega_s > \omega_M + \omega_c$, the range of sampling period is

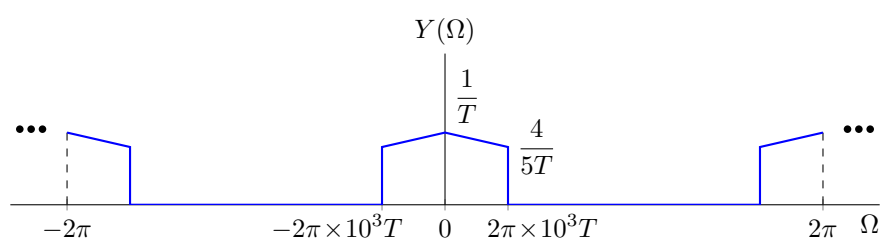
$$T = \frac{2\pi}{\omega_s} \implies \frac{1}{6}\times10^{-3} > T > \frac{1}{10}\times10^{-3} \text{ sec},$$

and the range of sampling frequency in Hz is

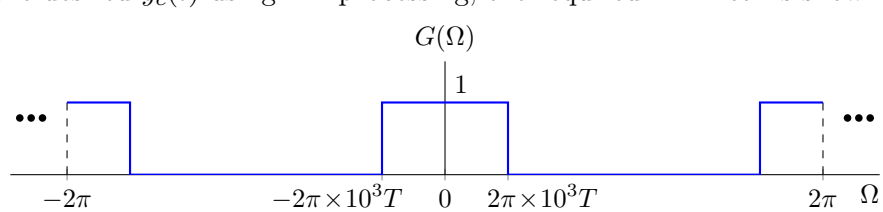
$$f_s = \frac{\omega_s}{2\pi} = \frac{1}{T} \implies 10\times10^3 > f_s > 6\times10^3 \text{ Hz}.$$

After sampling with the sub Nyquist rate, the spectrum of $x[n]$ and $y[n]$ are shown below.





To obtain the desired $y_c(t)$ using DT processing, the required DT filter is shown below.



When $\omega_s < \omega_M + \omega_c$, i.e. $f_s < 6$ kHz, $G(\Omega)$ cannot be expressed in terms of $H(\omega)$ to obtain $y_c(t)$, in other words DT processing of CT signals is not possible for this case.

6.	(a)	$$x[n] = \begin{cases} n, & -3 \le n \le 2 \\ 0, & \text{otherwise} \end{cases}$$ The given signal can be expressed as follows: $$x[n] = -3\delta[n+3] - 2\delta[n+2] - \delta[n+1] + \delta[n-1] + 2\delta[n-2]$$ The Fourier transform of the DT signal $x[n]$ is $$\begin{aligned} X(\Omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \\ &= -3e^{j3\Omega} - 2e^{j2\Omega} - e^{j\Omega} + e^{-j\Omega} + 2e^{-j2\Omega} \end{aligned}$$
	(b)	$$x[n] = \sin\left(\frac{\pi}{3}n\right) + \cos(n)$$ The given signal can be expressed as follows: $$x[n] = \frac{e^{j\frac{\pi}{3}n} - e^{-j\frac{\pi}{3}n}}{j2} + \frac{e^{jn} + e^{-jn}}{2}$$ The Fourier transform of the DT signal $x[n]$ is $$\begin{aligned} X(\Omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \\ &= \frac{\pi}{j}\left[\delta\left(\Omega - \frac{\pi}{3}\right) - \delta\left(\Omega + \frac{\pi}{3}\right)\right] + \pi\left[\delta\left(\Omega - 1\right) + \delta\left(\Omega + 1\right)\right], \qquad |\Omega| < \pi \end{aligned}$$
	(c)	$$x[n] = u[n-5] - u[n+1] + 8\delta[n-3]$$ The given signal can be expressed as follows: $$x[n] = -\delta[n+1] - \delta[n] - \delta[n-1] - \delta[n-2] + 7\delta[n-3] - \delta[n-4]$$ The Fourier transform of the DT signal $x[n]$ is $$\begin{aligned} X(\Omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \\ &= -e^{j\Omega} - 1 - e^{-j\Omega} - e^{-j2\Omega} + 7e^{-j3\Omega} - e^{-j4\Omega} \end{aligned}$$
	(d)	$$x[n] = \left(\frac{1}{3}\right)^{\lvert n\rvert} u[n+2]$$ The Fourier transform of the DT signal $x[n]$ is $$\begin{aligned} X(\Omega) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \\ &= \sum_{n=-2}^{\infty} \left(\frac{1}{3}\right)^{\lvert n\rvert} e^{-j\Omega n} \\ &= \frac{1}{9}e^{j2\Omega} + \frac{1}{3}e^{j\Omega} + \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^{n} e^{-j\Omega n} \\ &= \frac{1}{9}e^{j2\Omega} + \frac{1}{3}e^{j\Omega} + \frac{1}{1 - \frac{1}{3}e^{-j\Omega}} \end{aligned}$$

7. (a)

$$h[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi}{6}n\right) u[n] = \left(\frac{1}{3}\right)^n u[n] \cdot \sin\left(\frac{\pi}{6}n\right)$$

Fourier transform of each term is known and can be used directly.

$$h_1[n] = \left(\frac{1}{3}\right)^n u[n] \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad H_1(\Omega) = \frac{1}{1-\frac{1}{3}e^{-j\Omega}}$$

$$h_2[n] = \sin\left(\frac{\pi}{6}n\right) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad H_2(\Omega) = \frac{\pi}{j}\sum_{\ell=-\infty}^{\infty}\left[\delta\left(\Omega - \frac{\pi}{6} - 2\pi\ell\right) - \delta\left(\Omega + \frac{\pi}{6} - 2\pi\ell\right)\right]$$

Then, we can use the modulation (multiplication) property of DTFT.

$$h[n] = h_1[n]h_2[n] \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad H(\Omega) = \frac{1}{2\pi}H_1(\Omega) \circledast H_2(\Omega) = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} H_1(\Omega-\theta)H_2(\theta)\,\mathrm{d}\theta$$

$$\begin{aligned}
H(\Omega) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} H_1(\Omega-\theta)H_2(\theta)\,\mathrm{d}\theta \\
&= \frac{1}{j2}\int_{-\pi}^{\pi}\frac{1}{1-\frac{1}{3}e^{-j(\Omega-\theta)}}\underbrace{\sum_{\ell=-\infty}^{\infty}\left[\delta\left(\theta-\frac{\pi}{6}-2\pi\ell\right)-\delta\left(\theta+\frac{\pi}{6}-2\pi\ell\right)\right]}_{\delta\left(\theta-\frac{\pi}{6}\right)-\delta\left(\theta+\frac{\pi}{6}\right)\text{ for }\theta\in[-\pi,\pi]}\mathrm{d}\theta \\
&= \frac{1}{j2}\left(\frac{1}{1-\frac{1}{3}e^{-j\left(\Omega-\frac{\pi}{6}\right)}} - \frac{1}{1-\frac{1}{3}e^{-j\left(\Omega+\frac{\pi}{6}\right)}}\right) \\
&= \frac{1}{j2}\left(\frac{\frac{1}{3}e^{j\frac{\pi}{6}}e^{-j\Omega} - \frac{1}{3}e^{-j\frac{\pi}{6}}e^{-j\Omega}}{1-\frac{1}{3}e^{j\frac{\pi}{6}}e^{-j\Omega} - \frac{1}{3}e^{-j\frac{\pi}{6}}e^{-j\Omega} + \frac{1}{9}e^{-j2\Omega}}\right) \\
&= \frac{\frac{1}{3}\sin\left(\frac{\pi}{6}\right)e^{-j\Omega}}{1-\frac{2}{3}\cos\left(\frac{\pi}{6}\right)e^{-j\Omega}+\frac{1}{9}e^{-j2\Omega}}
\end{aligned}$$

The magnitude and phase of $H(\Omega)$ are given as below.

$$|H(\Omega)| = \sqrt{\left(\mathbf{Re}\left\{H(\Omega)\right\}\right)^2 + \left(\mathbf{Im}\left\{H(\Omega)\right\}\right)^2}$$

$$\angle H(\Omega) = \tan^{-1}\left(\frac{\mathbf{Im}\left\{H(\Omega)\right\}}{\mathbf{Re}\left\{H(\Omega)\right\}}\right)$$

We can also obtain the magnitude and phase of $H(\Omega)$ by separating it into functions of Ω.

$$H(\Omega) = \frac{\frac{1}{3}\sin\left(\frac{\pi}{6}\right)e^{-j\Omega}}{\left(1-\frac{1}{3}e^{-j\left(\Omega-\frac{\pi}{6}\right)}\right)\left(1-\frac{1}{3}e^{-j\left(\Omega+\frac{\pi}{6}\right)}\right)} = \frac{A(\Omega)}{B(\Omega)C(\Omega)}$$

where

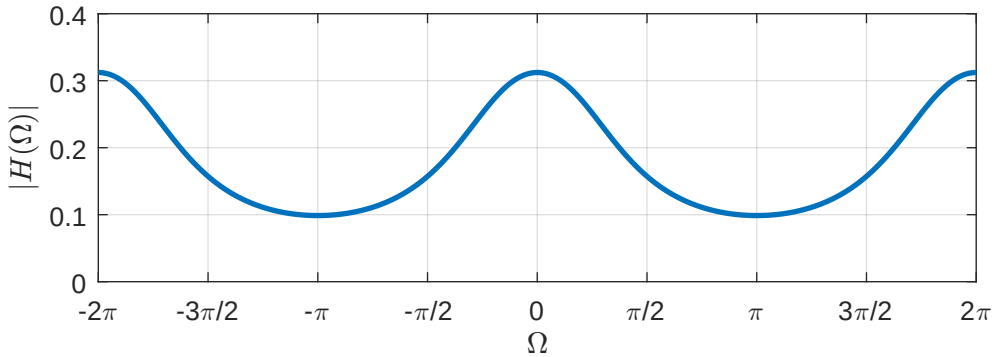
$$A(\Omega) = \frac{1}{3}\sin\left(\frac{\pi}{6}\right)e^{-j\Omega} = \frac{1}{6}\angle(-\Omega)$$

$$B(\Omega) = 1 - \frac{1}{3}e^{-j\left(\Omega-\frac{\pi}{6}\right)} = \sqrt{\frac{10}{9}-\frac{2}{3}\cos\left(\Omega-\frac{\pi}{6}\right)}\angle\tan^{-1}\left(\frac{\frac{1}{3}\sin\left(\Omega-\frac{\pi}{6}\right)}{1-\frac{1}{3}\cos\left(\Omega-\frac{\pi}{6}\right)}\right)$$

$$C(\Omega) = 1 - \frac{1}{3}e^{-j\left(\Omega+\frac{\pi}{6}\right)} = \sqrt{\frac{10}{9}-\frac{2}{3}\cos\left(\Omega+\frac{\pi}{6}\right)}\angle\tan^{-1}\left(\frac{\frac{1}{3}\sin\left(\Omega+\frac{\pi}{6}\right)}{1-\frac{1}{3}\cos\left(\Omega+\frac{\pi}{6}\right)}\right)$$

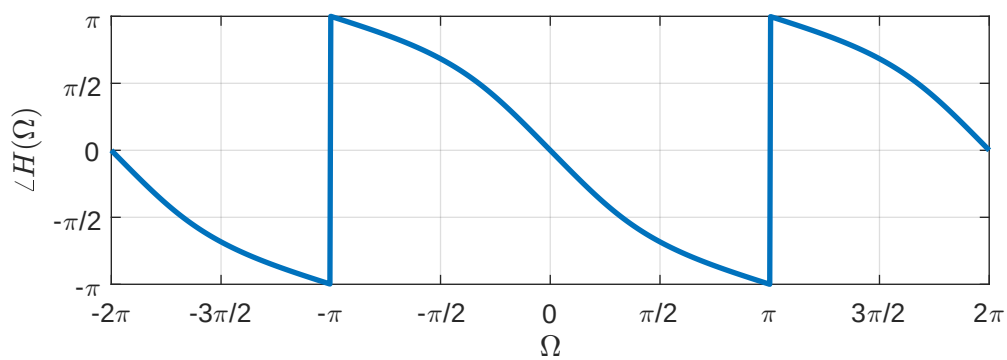
The magnitude of $H(\Omega)$:

$$|H(\Omega)| = \frac{|A(\Omega)|}{|B(\Omega)| \cdot |C(\Omega)|} = \frac{1}{6}\left[\left(\frac{10}{9} - \frac{2}{3}\cos\left(\Omega - \frac{\pi}{6}\right)\right)\left(\frac{10}{9} - \frac{2}{3}\cos\left(\Omega + \frac{\pi}{6}\right)\right)\right]^{-\frac{1}{2}}$$



The phase of $H(\Omega)$:

$$\begin{aligned}\angle H(\Omega) &= \angle A(\Omega) - \angle B(\Omega) - \angle C(\Omega) \\ &= -\Omega - \tan^{-1}\left(\frac{\frac{1}{3}\sin\left(\Omega - \frac{\pi}{6}\right)}{1 - \frac{1}{3}\cos\left(\Omega - \frac{\pi}{6}\right)}\right) - \tan^{-1}\left(\frac{\frac{1}{3}\sin\left(\Omega + \frac{\pi}{6}\right)}{1 - \frac{1}{3}\cos\left(\Omega + \frac{\pi}{6}\right)}\right)\end{aligned}$$



(b)

$$x[n] = (-1)^n \cos\left(\frac{\pi}{3}n\right)$$

Three possible ways to compute the system output $y[n]$ are

1. Convolution in time-domain:

$$y[n] = h[n] * x[n]$$

2. The convolution property of FT:

$$y[n] = \mathcal{F}^{-1}\{X(\Omega)H(\Omega)\} = \frac{1}{2\pi}\int_{\langle 2\pi\rangle} X(\Omega)H(\Omega)e^{j\Omega n}\,\mathrm{d}\Omega$$

3. Eigenfunction property for the input that is sum of complex exponentials:

$$y[n] = \sum_{k=1}^{K} H(\Omega_k) r_k\, e^{j\Omega_k n} \text{ for } x[n] = \sum_{k=1}^{K} r_k\, e^{j\Omega_k n}$$

The input $x[n]$ is the sum of complex exponentials. Hence, the third way is more convenient.

$$x[n] = (-1)^n \cos\left(\frac{\pi}{3}n\right) = e^{j\pi n}\left(\frac{e^{j\frac{\pi}{3}n} + e^{-j\frac{\pi}{3}n}}{2}\right) = \frac{1}{2}e^{j\frac{4\pi}{3}n} + \frac{1}{2}e^{j\frac{2\pi}{3}n}$$

$$y[n] = \frac{1}{2}\,H(4\pi/3)e^{j\frac{4\pi}{3}n} + \frac{1}{2}\,H(2\pi/3)e^{j\frac{2\pi}{3}n}$$

(c)

$$\begin{aligned}
h[n] &\overset{\mathcal{F}}{\longleftrightarrow} H(\Omega) \\
h[-n] &\overset{\mathcal{F}}{\longleftrightarrow} H(-\Omega) && \text{(Time reversal)} \\
h^*[-n] &\overset{\mathcal{F}}{\longleftrightarrow} H^*(\Omega) && \text{(Conjugation)} \\
g[n] = h[n] * h^*[-n] &\overset{\mathcal{F}}{\longleftrightarrow} G(\Omega) = |H(\Omega)|^2 = H(\Omega)H^*(\Omega) && \text{(Convolution)}
\end{aligned}$$

(d)

$$g[n] = e^{j\Omega_0 n}h[n] \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad G(\Omega) = H(\Omega - \Omega_0) \quad \text{(Frequency Shifting)}$$

8. (a)

$$h[n] = 3^n u[n] + \delta[n+4]$$

$h[n] \neq 0$ for $n < 0$, i.e. $h[-4] = 1 \implies h[n]$ is **not causal**.

$h[n]$ is not absolutely summable, i.e. $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$ does not hold $\implies h[n]$ is **not stable**.

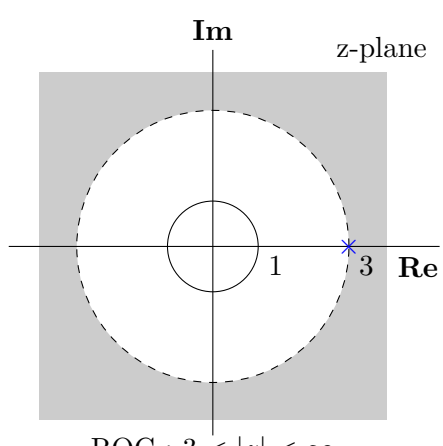
(b) The Fourier transform of $h[n]$

$$H(\Omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\Omega n} = \sum_{n=0}^{\infty} 3^n e^{-j\Omega n} + \sum_{n=-\infty}^{\infty} \delta[n+4]e^{-j\Omega n}$$

does **not converge** due to the first summation term ($3^n \to \infty$ as $n \to \infty$).

(c) The system (transfer) function

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n} = \sum_{n=0}^{\infty} 3^n z^{-n} + \sum_{n=-\infty}^{\infty} \delta[n+4]z^{-n} = \frac{1}{1-3z^{-1}} + z^4, \qquad 3 < |z| < \infty$$



ROC : $3 < |z| < \infty$

(d)	The ROC of $H(z)$ does not include the unit circle. Thus, the system is **not stable**. The ROC of $H(z)$ does not include ∞. Thus, the system is **not causal**. The answers are consistent with those in part (a).
(e)	The ROC of $H(z)$ does not include the unit circle. Thus, the DTFT of $h[n]$ $$H(\Omega) = H(z)\big\|_{z=e^{j\Omega}}$$ does **not converge**, as seen is part (b).
(f)	$\mathcal{F}\{r^{-n}h[n]\} = H(z)$ where $r = \|z\|$ Since $H(z)$ converges for $\|z\| > 3$, $\mathcal{F}\{r^{-n}h[n]\}$ converges for $r > 3$.
(g)	To express $x[n]$ in terms of $h[n]$, we utilize the given relation $H(z)\big\|_{z=5e^{j(\Omega+\pi)}} = \mathcal{F}\{x[n]\}$ as follows: $$\begin{aligned} H(z)\big\|_{z=5e^{j(\Omega+\pi)}} &= \sum_{n=-\infty}^{\infty} h[n]\left(5e^{j(\Omega+\pi)}\right)^{-n} \\ &= \sum_{n=-\infty}^{\infty} h[n]\left(5e^{j\pi}\right)^{-n} e^{-j\Omega n} \\ &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \\ &= \mathcal{F}\{x[n]\} \end{aligned}$$ Then, the desired sequence is $$x[n] = h[n]\left(5e^{j\pi}\right)^{-n} = (-5)^{-n}h[n] = (-0.6)^n u[n] + 5^4\delta[n+4].$$
(h)	$$(2j)^n h[-n] = \left(2e^{j\pi/2}\right)^n h[-n] = z_0^n h[-n]$$ $$\begin{aligned} h[n] &\overset{\mathcal{Z}}{\longleftrightarrow} H(z), && \text{ROC}: 3 < \|z\| < \infty \\ h[-n] &\overset{\mathcal{Z}}{\longleftrightarrow} H(z^{-1}), && \text{ROC}: 0 < \|z\| < \frac{1}{3} \quad \text{(Time reversal)} \\ z_0{}^n h[-n] &\overset{\mathcal{Z}}{\longleftrightarrow} H((z/z_0)^{-1}) && \text{ROC}: 0 < \|z\| < \frac{1}{3}\|z_0\| \quad \text{(Scaling in the z-domain)} \end{aligned}$$ $$g[n] = (2j)^n h[-n] \quad \overset{\mathcal{Z}}{\longleftrightarrow} \quad G(z) = H(2j/z) = \frac{1}{1-\frac{3}{2j}z} + 16z^{-4}, \qquad 0 < \|z\| < \frac{2}{3}$$